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SUPERSEDING
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MILITARY SPECIFICATION

LUMBER, HARDWOOD, MAHOGANY (FOR SHIP AND BOAT CONSTRUCTION)

This specification has been approved by the Department of Defense for use of the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 **Scope.**—This specification covers mahogany hardwood lumber of high quality suitable for use in ship and boat construction.

1.2 **Classification.**—The lumber shall be of the following classes, as specified (see 6.2):

Class PD—Planking and decking.
Class ST—Sawed timbers.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

TT-A-468 —Aluminum-Pigment; Powder and Paste, for Paint.

MILITARY

MIL-V-1174 —Varnish, Spar, Water-Resisting (Formula No. 80).
MIL-M-15176—Mica (Extender Pigment).

NAVY DEPARTMENT

General Specifications for Inspection of Material, and Appendix IV—Lumber and Timber.

STANDARDS

MILITARY

MIL-STD-129—Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.**—Material shall be American, African, or Philippine mahogany. American mahogany shall be of the species *Swietenia* macrophylla or *Swietenia* mahogoni grown in Mexico, the West Indies, Central America, and South America, as specified. (See 6.2.) African mahogany shall be of the species *Khaya* ivorensis or *Khaya* senegalensis grown in West Africa. Philippine mahogany shall be the dark red lauan species including *Shorea* polysperma (tangile), *Shorea* negrosensis, (red lauan), and *Shorea* teysmanniana (tiaong) grown in the Philippine Islands. (See 4.2.1 and 6.6 through 6.6.7.)

3.2 **Dimensions.**—Mahogany lumber shall be uniformly manufactured to not less than the dimensions specified (see 3.6.1, 6.2, and 6.5), with cross sections on the square and faces properly aligned.

3.2.1 Mahogany lumber shall have parallel edges and faces subject to the standard toler-

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ance for variations in sawing or subsequent surfacing. (See 4.3.1 and 6.2.)

3.3 **Defects.**—Except as hereinafter specified, mahogany lumber shall be free of the following defects: decay, sapwood, knots, holes, checks, cross breaks, streaks, excessive slope of grain, abnormal lightness, insect attack, excessive moisture, pith, splits, shakes, stain, wane, warp, chipped grain, loosened grain, raised grain, torn grain, skips, bark pockets, swirl figure and rough edges.

3.3.1 The defects shall be so scattered in any one piece that no combination of them at any one point is more weakening than a single maximum permissible defect.

3.3.2 *Slope of grain.*—Mahogany lumber shall not have a slope of grain, either diagonal or spiral or combination thereof, which exceeds 1 inch in 12 inches of length. (See 4.2.3.) Localized deviations around permitted defects will be admitted.

3.3.3 *Insect attack.*—The presence of frass or other indications of active insect attack will be cause for rejection.

3.3.4 *Lightness.*—Light weight, brash mahogany shall not be accepted. (See 3.5.)

3.4 **Moisture content.**—Mahogany lumber shall be seasoned by air or kiln drying to be within the following moisture content limits: (See 4.4.1.)

Class PD—15 ± 3 percent.

Class ST—Any stage of seasoning.

3.5 **Specific gravity (density).**—Exceptionally lightweight mahogany lumber shall be rejected. (See 6.4.) Mahogany lumber suspected of being abnormally light in weight during inspection handling shall be tested (see 4.4.2), if necessary. The specific gravity of mahogany lumber so tested shall not be less than 0.45 (28.1 pounds per cubic foot density).

3.6 **Class PD, planking and decking.**—Wane or its equivalent in pin wormholes along an edge will be admitted in the aggregate of $\frac{1}{12}$ of the surface measurement. One sound knot, excluding pin knots, not exceeding ½ inch in diameter, will be admitted in each 3 square feet of surface if 1½ feet or more apart. Scattered pin knots will be admitted. Warp and cup will be admitted if not too great to prevent the board from being dressed on two sides to standard thickness. Closed gum pockets less than 1½ inches long will be admitted in the same number and amount as ½ inch knots. End checks not over $\frac{1}{32}$ inch wide and 4 inches long and surface checks of perceptible opening not over 2½ inches long will be admitted. Medium and large surface checks will not be admitted. Sapwood will be admitted in an amount no greater than ⅛ the width of any surface on which it appears.

3.6.1 Width, for all thicknesses, shall be 6 inches and over (average not less than 9 inches), and length shall be 8 feet and over (average not less than 14 feet).

3.7 **Class ST, sawed timber.**—Wane will be admitted on timber 12 inches or more in the greatest dimension. Wane $\frac{1}{12}$ the width of the timber will be admitted, only on one or both edges on the top face. One sound knot not exceeding ¾ inch in diameter, or its equivalent in pin wormholes or closed gum or bark pockets, will be admitted in each 4 square feet of surface if 2 feet or more apart. Scattered pin knots will be admitted. End checks not over $\frac{1}{32}$ inch wide and 6 inches long and medium surface checks will be admitted. Sapwood will be admitted in an amount no greater than ⅛ the width of any surface on which it appears.

3.8 **Coatings.**—The ends of sawn timber shall be painted with 2 coats of aluminum or mica paint, the first coat to be applied within 24 hours after sawing. Aluminum paint shall consist of 2 pounds of aluminum powder paste conforming to type II, class A of Specification TT–A–468 and 1 gallon of phenolic

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varnish conforming to Specification MIL–V–1174. Mica paint shall consist of 2 pounds of mica conforming to type I of Specification MIL–M–15176, and 1 gallon of phenolic varnish conforming to Specification MIL–V–1174. A drying interval of 24 hours shall elapse before applying a second coat. A second alternate end coating may be any commercial end coating satisfactory to the bureau or agency concerned.

3.8.1 *Side coatings.*—When decking, planking, and sawn timber are required to be side coated (see 6.2), they shall be given two brush or spray applications of aluminum paint consisting of 2 pounds of aluminum paste conforming to type II, class B of Specification TT–A–468 and 1 gallon of phenolic varnish conforming to Specification MIL–V–1174. The side coating shall be applied before shipment with a drying interval of 24 hours before applying the second coat. Dipping in fungicides, when required (see 5.1), shall precede side coating applications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection lot.—All mahogany of one grade which is offered for delivery at one time and will be transported in one truck or railroad car shall be considered a lot for purposes of inspection.

4.1.1 *Quantity.*—The quantity of lumber in the lot shall be determined by tally, in accordance with commercial practice as outlined in applicable portions of General Specifications for Inspection of Material, Appendix IV, Part III.

4.1.2 *Sampling for dimensional inspection and moisture content.*—Sample pieces of lumber shall be selected in accordance with table I for the inspection and tests specified in 4.3 and 4.4.1.

4.2 Inspection.—Each piece of mahogany shall be visually inspected by the Government inspector to determine compliance with the requirements of this specification for species, grade,[1] and absence of insect infestation.

TABLE I.—*Sampling for dimensional inspection and moisture content A.Q.L. = 4.0 percent (approx.) nonconforming.*

Lot size, pieces of lumber	Sample size, pieces of lumber	Acceptance number defective	Rejection number defective
Up to 110	15	1	2
111 to 180	25	2	3
181 to 500	35	3	4
501 to 800	50	4	5
801 to 1,300	75	6	7
1,301 to 3,200	110	8	9
3,201 and over	150	11	12

4.2.1 **Species.**—The Government inspector is authorized to submit samples of lumber of doubtful species to a laboratory designated by the bureau or agency concerned for positive identification. (See 6.6 through 6.6.7.)

4.2.2 *Inspection procedure.*—The Government inspector shall be present during the loading of the lumber into the highway truck or railroad car and shall, as the 100 percent grade inspection is carried out, select sample pieces at random as the lumber is moved, for purposes of moisture content and dimensional determination. The number of pieces in lot shall be tallied, and the sample required by table I shall be selected. Each sample piece shall be examined only for conformance to the requirements of this specification for dimensions and moisture content specified. Any sample piece which does not meet the requirements shall be rejected, and if the number of such pieces found in any sample exceeds the acceptance number in table I for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again after the contractor has removed all nonconforming pieces.

[1] Excessive sapwood, slope of grain and lightness (see 3.3) are classified as grade defects. Slope of grain and lightness shall be determined during the 100 percent visual inspection and questionable pieces shall be tested in accordance with 4.2.3 and 4.4.2, if necessary.

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4.2.3 *Slope of grain.*—Slope of grain shall be determined on two adjoining sides of lumber suspected of having spiral or diagonal grain, or combination thereof.

4.3 **Measurement.**—Measurement of the diameter of knots or holes shall be determined by averaging the distance between lines drawn parallel to the edges and ends of the piece and touching the extremities of the knot or hole being measured.

4.3.1 *Dimensions.* — Dimensions of both rough and surfaced lumber shall be subject to the standard tolerances specified for hardwood lumber in General Specifications for Inspection of Material, Appendix IV.

4.4 **Test procedures.**

4.4.1 *Moisture content.*—Moisture content shall be determined in accordance with the procedures outlined in Appendix IV of the General Specifications for Inspection of Material, or by a moisture meter approved by the bureau or agency concerned.

4.4.2 *Specific gravity.*—The specific gravity shall be determined by ascertaining the ratio of the volume below the waterline (see fig. 1) to that of the whole piece. The determination of specific gravity by this method is done as follows:

(a) Cut a piece of wood from the lumber to be tested, about 1 by 1 by 12 inches in dimension, the 12-inch dimension being along the grain.

(b) Dry the piece for 48 hours at 212° to 221° F. heat.

(c) Dress the piece so that the cross section is rectangular and uniform throughout the length, and trim it to exactly 10 inches in length. Mark off and number the length in one inch demarcations.

(d) Place the piece carefully in a tall glass container of water so that it floats in an upright position. Note the water line to which it sinks. Quickly remove and mark the average wetted length (see fig. 1). The position of the water line indicates the specific gravity of the piece on the basis of oven-dry weight and oven-dry volume. Thus, if the mark is halfway between 4 and 5 inches the specific gravity is 0.45. The whole operation must be performed quickly, because the oven-dry piece will begin to pick up moisture while it is being dressed, cut, and marked and when it is placed in the water. When determining the acceptability of suspected material the required immersion depth may be marked on the specimens before testing.

5. PREPARATION FOR DELIVERY

5.1 **Stain and mold protection of the green class ST lumber.**—When class ST mahogany lumber is shipped "green" during warm weather or when the shipment will be in transit for more than 15 days precautions shall be taken to prevent the development of stain or decay causal fungi. The preventive treatment shall consist of dipping in or spraying with a water solution of 22½ pounds of sodium pentachlorophenate in 100 gallons of water, or of ethylmercuric phosphate in a solution containing 6 ounces of the active ingredient in 100 gallons of water, or other stain preventing water solutions of equal effectiveness.

5.2 **Packaging.**—Narrow, thin pieces shall be securely bundled to facilitate handling.

5.3 **Packing.**—Unless otherwise specified in the contract or order, the loading of the lumber shall be as directed by the Government inspector. As far as practicable the lumber shall be segregated for class, grade, thick-

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ness and width. The load shall be blocked and braced to prevent shifting in transit and shall be protected against precipitation and the adverse effects of the direct rays of the sun in shipment. Surfaced lumber and dried lumber shall be shipped only in closed cars or vans and shall be stored under cover at all times.

5.4 **Marking.**—In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Standard MIL–STD–129 for carload shipments. Suitable tally cards shall be conspicuously posted at access entranceways within the carrier. The tally cards shall be protected against deterioration and loss.

6. NOTES

6.1 **Intended use.**—This material is intended for use as solid timbers in small boats and for solid planking of patrol craft, and other wooden boats and ships.

6.2 **Ordering data.** — Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Species (African, American, or Philippine).
(d) Thickness, width, and length required (see 3.2).
(e) The amount of surfacing desired, if any (see 3.2.1).
(f) Whether side coatings are required (see 3.8.1).
(g) That the contractor shall pile material awaiting shipment on stickers, and protect it from precipitation and the direct rays of the sun, when directed by the Government inspector.

6.3 This specification specifies the extent of defects and sapwood (see 6.6) permitted in the poorest pieces of timber acceptable.

6.4 Exceptionally lightweight pieces of mahogany lumber have decidedly decreased strength properties. Suspected lightweight lumber should be tested, if necessary.

6.5 Because of the nonconformance to standard commercial dimensions, particularly sawn timber, of the mahogany specified in this specification, it is usually necessary to manufacture it to order. The use of side and end coatings as specified in 3.8 and 3.8.1 reduces the amount of seasoning degrade, and should be specified in all requests for mahogany for use by West Coast activities and activities outside the limits of the continental United States. Side coatings are not necessary on mahogany which has been thoroughly seasoned before shipment.

6.6 It is important that mahogany lumber secured under this specification be of the correct species. Many inferior species of tropical hardwoods frequently are sold as "mahogany". Such species usually have reduced strength properties, less natural decay resistance, show poor working qualities and frequently are quite similar in appearance to the species of mahogany required. Use of such material for building wooden craft requires added labor during construction, and results in increased costs during service brought about by replacement of expensive wood components broken due to inadequate strength or decayed because of insufficient natural decay-resistance.

6.6.1 The woods commonly sold in the United States as mahogany, with or without a qualifying term indicating their origin, may be classed in three commercial groups: (a) woods from tropical America consisting of several species of the botanical genus Swietenia; (b) a number of species of the African genus *Khaya*, termed "khaya" and frequently called "African mahogany" in the trade; and (c) various Philippine hardwoods, principally tangile and red lauan, both of which belong to the botanical genus *Shorea*, often called "Philippine mahogany" in the trade.

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6.6.2 The color of all these woods comprises various shades of reddish brown. Tangile, however, which constitutes the bulk of "Philippine mahogany," and some species of khaya, often have a purplish tinge. In mahogany, and to a less extent in khaya, the color darkens with prolonged exposure to light, whereas unstained tangile becomes lighter colored. To observe the original color, therefore, fresh surfaces should be exposed.

6.6.3 All the woods in these three groups usually have interlocked grain and are lustrous, which gives quarter-sawed lumber and radially-cut veneer a pronounced ribbon figure, that is, alternating dark and light stripes, an inch, more or less, in width. In all of them the pores can be seen without a magnifying glass as minute holes on smoothly cut end surfaces and as grooves of varying lengths on planed longitudinal surfaces. The pores are fairly uniform in size and fairly evenly distributed on all surfaces.

6.6.4 Wood of a light grey or pale brown color is probably not of the species permitted.

6.6.5 *Swietenia* and *khaya* have a distinctive characteristic in common, namely, the occurrence of dark amber-colored gum in many of the pores. This gum can be seen on planed or split longitudinal surfaces without a magnifying glass; but better with a glass, and it distinguishes *Swietenia* and *khaya* from *Shorea*, in which such gum does not occur.

6.6.6 *Swietenia* and *khaya* can be distinguished from each other by the presence in the former and absence in the latter of fine, continuous light-colored more or less parallel lines $\frac{1}{32}$ to $\frac{1}{2}$ inch apart on smoothly cut end-grain and quarter-sawed surfaces, and slightly to considerably farther apart on plain-sawed surfaces. Swietenia frequently contains a white deposit in the pores which is absent in khaya.

6.6.7 In *Shorea* species the pores are partly filled with iridescent, froth-like growths known as "tyloses". On smoothly cut end surfaces, *Shorea* species show more or less broken parallel lines one-eighth inch to several inches apart, readily visible to the naked eye, and appearing under a lens as rows of minute openings, smaller than the pores, filled with a white substance.

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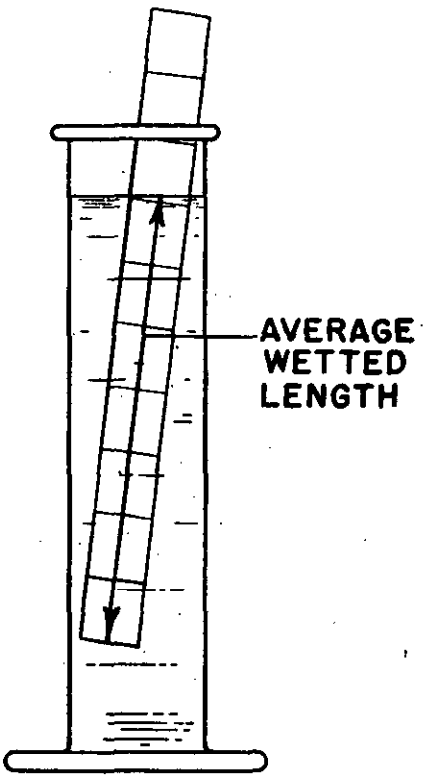


FIGURE 1.—*Specific gravity.*

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